

1.3 Computer networks, connections and protocols

How to run this unit, lesson by lesson

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Start here

15 lessons. A realistic estimate is **15 teaching hours**: most lessons are one hour, and a lesson with thirty or more activities takes two or three. The estimate counts transitions and assumes you teach from the front for the first fifteen minutes of each lesson.

Teach them in the order below. The order is the retrieval schedule as well as the teaching one: each starter asks about the lesson before it.

#	LESSON	HOURS	PLAN
1	Types of network: zoom out nw-101	1	Lesson_Plans/NW_L01_LessonPlan.pdf
2	Network control room nw-102	1	Lesson_Plans/NW_L02_LessonPlan.pdf
3	Network showdown nw-103	1	Lesson_Plans/NW_L03_LessonPlan.pdf
4	Mission: wire up the school nw-104	1	Lesson_Plans/NW_L04_LessonPlan.pdf
5	Inside the internet nw-105	1	Lesson_Plans/NW_L05_LessonPlan.pdf
6	Star and mesh networks nw-106	1	Lesson_Plans/NW_L06_LessonPlan.pdf
7	Revision HQ nw-107	1	Lesson_Plans/NW_L07_LessonPlan.pdf
8	Assessment 1 nw-108-test	1	Lesson_Plans/NW_L08_TEST_LessonPlan.pdf
9	Connection café nw-109	1	Lesson_Plans/NW_L09_LessonPlan.pdf
10	Cipher room nw-110	1	Lesson_Plans/NW_L10_LessonPlan.pdf
11	Address sorting office nw-111	1	Lesson_Plans/NW_L11_LessonPlan.pdf
12	Protocol city nw-112	1	Lesson_Plans/NW_L12_LessonPlan.pdf
13	Parcel depot nw-113	1	Lesson_Plans/NW_L13_LessonPlan.pdf
14	Assessment 2 reflection nw-114-test	1	Lesson_Plans/NW_L14_TEST_LessonPlan.pdf
14	Revision HQ 2 nw-114	1	Lesson_Plans/NW_L14_LessonPlan.pdf

Preparation and what you need

1. **Class logins, before the first lesson.** On the teacher dashboard ([teacher.html](#)), open **Class logins**, enter the class and the usernames one per line, and print the login cards. A pupil cannot sign in without one.
2. **Devices.** A laptop, a tablet or a phone will do. A headset is optional everywhere. The browser needs to be current; the experiences use WebGL.
3. **Print the worksheets** for each lesson. They are in this pack as .docx to edit and .pdf to print.

4. **Open each experience once yourself** on the teaching machine, so the images are cached before thirty pupils ask for them at the same moment.

Where progress is kept. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device a pupil used, not on a server. The teacher dashboard says so itself. Keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels with the pupil.

Pupils sign in at topics.html with the username and PIN on their card, choose the topic and then the lesson.

Using the three resources together

Slides for the front of the room; **the experience** for reading, deciding and being marked; **the worksheet** for writing and for the record. The rule that holds them together is write first, then answer at the badge.

Do not teach the station walls and then send pupils to read the same walls. Teach the first two or three from the slides and let them read the rest; the walls are there so a pupil can re-read what you said, not so you can say it twice.

Evidence, checking and common difficulties

Collect three things from every lesson: the starter (written), the key facts and station challenges on the worksheet (written), and the score screen breakdown (copied onto the worksheet by the pupil as soon as it appears). The score screen shows the marks per station, which is what tells you where to reteach; a single total does not.

The teacher dashboard shows progress by pupil and the class's weakest stations, averaged on first attempts. Use the weakest-station list to choose what to reteach rather than asking who found it hard.

Things to plan around in this unit

- only one activity asks the pupil to produce an answer rather than choose one — nw-101, nw-102, nw-103, nw-104, nw-107, nw-110, nw-111, nw-112, nw-114
- no optional challenge activity — nw-101, nw-102, nw-103, nw-104, nw-105, nw-106, nw-107, nw-109, nw-110, nw-111, nw-112, nw-113, nw-114

The lesson plans, in teaching order

Each lesson has its own plan as a separate PDF in Lesson_Plans/ of this pack, so you can print the one you are teaching. What follows is the summary of each; the plan carries the timings, the hinge question, the station-by-station delivery and the adaptations.

1. Types of network: zoom out nw-101

What do you gain, and what do you give up, by connecting a computer to a network?

OUTCOMES	know what is meant by a standalone computer; know the different types of network: LAN and WAN; understand the advantages of networking; understand the implications (disadvantages) of networking
STATIONS AND ACTIVITIES	15 stations, 27 activities, 35 marks (26 guided, 1 independent, 0 challenge)
RESOURCES	NW_L01_TypesOfNetworkZoomOut_Lesson.pptx, NW_L01_TypesOfNetworkZoomOut_Worksheet.docx
PLAN	Lesson_Plans/NW_L01_LessonPlan.pdf

2. Network control room nw-102

Why is the same network fast at one moment and slow at another?

OUTCOMES	understand what factors can affect the performance of a network; be able to explain how each factor affects network speed
STATIONS AND ACTIVITIES	6 stations, 14 activities, 23 marks (13 guided, 1 independent, 0 challenge)
RESOURCES	NW_L02_NetworkControlRoom_Lesson.pptx, NW_L02_NetworkControlRoom_Worksheet.docx
PLAN	Lesson_Plans/NW_L02_LessonPlan.pdf

3. Network showdown nw-103

What are the differences between peer-to-peer and client-server networks?

OUTCOMES	know what a client-server model is; know what a peer-to-peer model is; understand the different roles computers have in each model
STATIONS AND ACTIVITIES	6 stations, 13 activities, 26 marks (12 guided, 1 independent, 0 challenge)
RESOURCES	NW_L03_NetworkShowdown_Lesson.pptx, NW_L03_NetworkShowdown_Worksheet.docx
PLAN	Lesson_Plans/NW_L03_LessonPlan.pdf

4. Mission: wire up the school nw-104

What hardware does it take to turn a room full of computers into a network?

OUTCOMES	know the hardware needed to connect a LAN; understand the purpose of each piece of hardware and the jobs it does
STATIONS AND ACTIVITIES	6 stations, 13 activities, 16 marks (12 guided, 1 independent, 0 challenge)
RESOURCES	NW_L04_MissionWireUpTheSchool_Lesson.pptx, NW_L04_MissionWireUpTheSchool_Worksheet.docx
PLAN	Lesson_Plans/NW_L04_LessonPlan.pdf

5. Inside the internet nw-105

How does the internet work?

OUTCOMES	understand what the internet actually is; understand the terms Domain Name Service and domain name server; understand what is meant by hosting and the cloud; understand what is meant by the terms web server and client
STATIONS AND ACTIVITIES	6 stations, 13 activities, 27 marks (11 guided, 2 independent, 0 challenge)
RESOURCES	NW_L05_InsideTheInternet_Lesson.pptx, NW_L05_InsideTheInternet_Worksheet.docx
PLAN	Lesson_Plans/NW_L05_LessonPlan.pdf

6. Star and mesh networks nw-106

Which shape of network would you fit here, and what are you giving up by choosing it?

OUTCOMES	know what a star network is; know what a mesh network is; understand that the internet is an example of a partial mesh network; know the advantages and disadvantages of star and mesh networks
STATIONS AND ACTIVITIES	10 stations, 22 activities, 32 marks (20 guided, 2 independent, 0 challenge)
RESOURCES	NW_L06_StarAndMeshNetworks_Lesson.pptx, NW_L06_StarAndMeshNetworks_Worksheet.docx
PLAN	Lesson_Plans/NW_L06_LessonPlan.pdf

7. Revision HQ nw-107

Which parts of this topic could I explain to somebody else, and which can I only recognise when I see them?

OUTCOMES	know which lessons I am secure on and which I need to revise; have completed any outstanding work from lessons 1 to 6; be ready for my mid-topic test next lesson
STATIONS AND ACTIVITIES	6 stations, 19 activities, 23 marks (18 guided, 1 independent, 0 challenge)
RESOURCES	NW_L07_RevisionHQ_Lesson.pptx, NW_L07_RevisionHQ_Worksheet.docx
PLAN	Lesson_Plans/NW_L07_LessonPlan.pdf

8. Assessment 1 nw-108-test

Your 1.3 test on lessons 1 to 7 is done on paper. Warm up first with Revision HQ, then use this reflection sheet once it's marked.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.
STATIONS AND ACTIVITIES	0 stations, 0 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	NW_L08_Assessment1_Reflection_Worksheet.docx
PLAN	Lesson_Plans/NW_L08_TEST_LessonPlan.pdf

9. Connection café nw-109

Which is better: a wired or wireless network?

OUTCOMES	understand that Ethernet is a wired method of connection; understand that Wi-Fi and Bluetooth are wireless methods of connection; understand the benefits and drawbacks of wired versus wireless connections; be able to recommend a connection type for a specified scenario
STATIONS AND ACTIVITIES	6 stations, 13 activities, 28 marks (11 guided, 2 independent, 0 challenge)
RESOURCES	NW_L09_ConnectionCafe_Lesson.pptx, NW_L09_ConnectionCafe_Worksheet.docx
PLAN	Lesson_Plans/NW_L09_LessonPlan.pdf

10. Cipher room nw-110

What is the purpose of encryption?

OUTCOMES	know the basics of how cryptography can work with a simple key; know how wireless devices authenticate each other before communicating data; understand the difference between private and public keys; understand why private keys are never shared
STATIONS AND ACTIVITIES	6 stations, 13 activities, 24 marks (12 guided, 1 independent, 0 challenge)
RESOURCES	NW_L10_CipherRoom_Lesson.pptx, NW_L10_CipherRoom_Worksheet.docx
PLAN	Lesson_Plans/NW_L10_LessonPlan.pdf

11. Address sorting office nw-111

What are the differences between the three types of network address?

OUTCOMES	understand the uses of MAC and IP addressing; understand the difference between IPv4 and IPv6; understand the need for IPv6
STATIONS AND ACTIVITIES	6 stations, 13 activities, 21 marks (12 guided, 1 independent, 0 challenge)
RESOURCES	NW_L11_AddressSortingOffice_Lesson.pptx, NW_L11_AddressSortingOffice_Worksheet.docx
PLAN	Lesson_Plans/NW_L11_LessonPlan.pdf

12. Protocol city nw-112

What are standards and protocols?

OUTCOMES	understand the need for standards in computing; understand the seven common protocols and what they are used for
STATIONS AND ACTIVITIES	6 stations, 14 activities, 28 marks (13 guided, 1 independent, 0 challenge)
RESOURCES	NW_L12_ProtocolCity_Lesson.pptx, NW_L12_ProtocolCity_Worksheet.docx
PLAN	Lesson_Plans/NW_L12_LessonPlan.pdf

13. Parcel depot nw-113

What are the benefits of layering protocols?

OUTCOMES	know why protocols are layered; understand the advantages of layering
STATIONS AND ACTIVITIES	6 stations, 12 activities, 25 marks (10 guided, 2 independent, 0 challenge)
RESOURCES	NW_L13_ParcelDepot_Lesson.pptx, NW_L13_ParcelDepot_Worksheet.docx
PLAN	Lesson_Plans/NW_L13_LessonPlan.pdf

14. Assessment 2 reflection nw-114-test

Once your end-of-unit test is marked, use this sheet to plan what to improve.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.
STATIONS AND ACTIVITIES	0 stations, 0 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	NW_L14_Assessment2_Reflection_Worksheet.docx
PLAN	Lesson_Plans/NW_L14_TEST_LessonPlan.pdf

14. Revision HQ 2 nw-114

Which parts of this topic could I explain to somebody else, and which can I only recognise?

OUTCOMES	know which lessons I am secure on and which to revise; be ready for the end-of-unit assessment
STATIONS AND ACTIVITIES	6 stations, 20 activities, 30 marks (19 guided, 1 independent, 0 challenge)
RESOURCES	NW_L14_RevisionHQ2_Lesson.pptx, NW_L14_RevisionHQ2_Worksheet.docx
PLAN	Lesson_Plans/NW_L14_LessonPlan.pdf

Teacher answers

Model answers and marking guidance are not in this guide and not on the lesson plans. A plan is printed and left on a desk; an answer sheet is not. Where verified answers exist for this unit they are in the Teacher_Answers/ folder of this pack. Where they do not, the pack says so rather than shipping an empty file.

For the programming lessons there is no single model answer to print: a program is marked by running it against its test cases, and many different correct programs pass. The tests themselves are the marking guidance, and they are in the teacher answers.

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